

MA108T: Calculus

Quiz one

Instructor: Shraddha Srivastava

Date: 26/08/2025

Duration: 2:00 to 3:00 pm

Maximum marks: 15

Guidelines:

1. Calculators, mobile phones, smart watch, etc electronic devices, notes, books, and cheat sheets are **not** allowed.
2. Kindly answer all the questions. You may answer them in any order. Please read your questions carefully.
3. If you feel that a question is incorrect, please write your claim with proper justification- if your claim is found to be correct, you will get full marks for the corresponding question.
4. If you are done answering all questions, you may leave the examination hall **silently**.

Question 1: [2 marks] State **true** or **false**.

- (i) Every bounded sequence is convergent.
- (ii) Every continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ is bounded.
- (iii) Every bounded function is continuous.
- (iv) Let $(x_n)_{n \in \mathbb{N}}$ be sequence such that subsequences $(x_{2n})_{n \in \mathbb{N}}$ and $(x_{2n+1})_{n \in \mathbb{N}}$ are convergent. Then $(x_n)_{n \in \mathbb{N}}$ is convergent as well.

Question 2: [2 marks] Prove that the sequence $x_n := (1 + \frac{1}{n})^n$ is convergent.

Question 3: [2 marks] Find the supremum and infimum of the set $\{\frac{m}{m+n} \mid m, n \in \mathbb{N}\}$. **Prove your claim.**

Question 4: [2 marks] Using $(\epsilon - n_0)$ -definition prove that the sequence $x_n := \frac{n^2+2n+100}{n^3+1}$ is convergent.

Question 5: [3 marks] Using $(\epsilon - \delta)$ -definition prove that the function $f(x) = x^3 + 2x - 1$ is continuous at 1.

Question 6: [2 marks] Is the following function

$$f(x) = \begin{cases} x \cos(\frac{1}{x}), & \text{if } x \neq 0; \\ 0, & \text{if } x = 0; \end{cases}$$

differentiable? **Prove your claim.**

Question 7: [2 marks] Let $f(x) = x^3 - x^2 + x$. Using IVT show that there is $c > 2$ such that $f(c) = 10$.

MA108T: Calculus

Mid semester exam

Instructor: Shraddha Srivastava

Date: 28/09/2025

Duration: 9:30 to 11:30 am

Maximum marks: 30

Guidelines:

1. Calculators, mobile phones, smart watch, etc electronic devices, notes, books, and cheat sheets are **not** allowed.
2. Kindly answer all the questions. You may answer them in any order. Please read your questions carefully.
3. If you feel that a question is incorrect, please write your claim with a proper justification - if your claim is found to be correct, you will get full marks for the corresponding question.

Notation: $\mathbb{R}$:= the set of real numbers, $\mathbb{N}$:= the set of natural numbers, $\mathbb{Q}$:= the set of the rational numbers.

Question 1: [5 marks] State **true** or **false**. Justify your answer.

- (i) There exists a continuous function $f : [1, 2] \rightarrow \mathbb{R}$ whose range is $[0, \infty)$.
- (ii) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f' = 0$. Then f must be constant.
- (iii) The sequence $(\frac{2^n}{n!})$ is convergent.

Question 2:

- (i) [1 mark] Let $\frac{a}{b} \in \mathbb{Q}$. Construct a sequence of irrational numbers which converges to $\frac{a}{b}$.
- (ii) [2 marks] Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function such that $f(x) = 0$ for each **irrational** number x . Using sequential criteria for continuity show that $f(y) = 0$ for all $y \in \mathbb{R}$.

Question 3: [3 marks] Using $(\epsilon - n_0)$ -definition prove that the sequence with its n th term

$$x_n := \frac{n^2 + 2n - 5}{n^5 + 6n + 7}$$

is convergent.

Question 4: [4 marks] Using $(\epsilon - \delta)$ -definition prove that the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are continuous at the respective a

- (i) $a = -2$ and $f(x) = 3x^2 + 7x + 2$
- (ii) $a = 0$ and $f(x) = \begin{cases} x^4 \sin(\frac{1}{x^3}), & \text{if } x \neq 0; \\ 0, & \text{if } x = 0. \end{cases}$

Please turn over!

Question 5: [3 marks] Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at $c \in \mathbb{R}$ such that $f(c) > 0$. Prove that there exists $r > 0$ such that $f(x) > 0$ for all $x \in (c - r, c + r)$.

Question 6: [2 marks] Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(x + y) = f(x) + f(y)$ and f is differentiable at 0. **Prove or disprove** that f is differentiable at any point arbitrary of $\mathbb{R}$.

Question 7: [2 marks] Find the intervals of increase/decrease and convex/concave for

$$f(x) = 2x^3 + 2x^2 - 2x - 1.$$

Question 8: [4 marks] State Taylor's theorem and apply it to prove the following.

(i) For $x > 0$, $1 + \frac{x}{2} - \frac{x^2}{8} < \sqrt{1+x} < 1 + \frac{x}{2}$.

(ii) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f''(x) \geq 0$ for all $x \in \mathbb{R}$. Further you may assume that there is x_0 such that $f'(x_0) \geq 0$. Show that if f is bounded, then f must be constant.

Question 9: [4 marks] Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Give the definition of f being a Riemann integrable function. Is the following function $h : [a, b] \rightarrow \mathbb{R}$ Riemann integrable?

$$h(x) = \begin{cases} 0, & \text{if } a \leq x < b; \\ \sqrt{2}, & \text{if } x = b. \end{cases}$$

If yes, find $\int_a^b h(x) dx$ (in particular, you need to find $U(h)$ and $L(h)$) using the definition of Riemann integration.

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Quiz two

Instructor: Shraddha Srivastava

Date: 28/10/2025

Duration: 2:00 to 3:00 pm

Maximum marks: 20

Guidelines:

1. Calculators, mobile phones, smart watch, etc electronic devices, notes, books, and cheat sheets are **not** allowed.
2. Kindly answer all the questions. You may answer them in any order. Please read your questions carefully.
3. If you feel that a question is incorrect, please write your claim with proper justification- if your claim is found to be correct, you will get full marks for the corresponding question.
4. If you are done answering all questions, you may leave the examination hall **silently**.

Question 1: [4 marks] Find the first order partial derivatives of the following functions:

(a) $f(x, y) = e^{xy}$ (b) $g(x, y, z) = x \sin(x + y) + z$

Question 2: [3.5 marks] Using Lagrange's multiplier method, find the maximum and minimum of the function $f(x, y) = xy$ on the ellipse whose equation is $4x^2 + y^2 = 8$.

Question 3: Do the following:

- (i) [1.5 marks] Check whether the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^4}{x^7 + y^6}$ exists or not. Justify your answer.
- (ii) [2 marks] Find the maximum, minimum rate of change of $f(x, y) = x^4 y - x^2 y^3$ at $(2, -3)$ and the unit vectors in whose directions these occur.

Question 4: Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{x^2 y^2 \sin(\frac{1}{y})}{\sqrt{x^4 + y^2}}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

- (i) [2.5 marks] Is f continuous at $(0, 0)$? If yes, prove your claim using $\epsilon - \delta$ definition.
- (ii) [3.5 marks] Is f differentiable at $(0, 0)$? If yes, prove it using the definition of the differentiability of f at point given in the class.

Question 5: [3 marks] Suppose that $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a differentiable function. Suppose that the curve

$$C := \{(x(t), y(t)) \mid t \in \mathbb{R}\}$$

is the level curve $f(x, y) = k$ for some $k \in \mathbb{R}$, where $x(t)$ and $y(t)$ are differentiable functions of t . Let $(x_0, y_0) \in C$. State and use an appropriate chain rule to prove that the gradient vector $\nabla f(x_0, y_0)$ is perpendicular to the level curve $f(x, y) = c$.

MA108T: Calculus

End semester exam

Instructor: Shraddha Srivastava

Date: 23/11/2025

Duration: 9:30 am to 12:30 pm

Maximum marks: 50

Guidelines:

1. Calculators, mobile phones, smart watches and such electronic devices, notes, books, and cheat sheets are **not** allowed.
2. Kindly answer all the questions. You may answer them in any order. Please read your questions carefully.
3. If the exact calculation appears in more than one questions, then you need not repeat it.

Question 1: [2 marks] Does there exist a differentiable function $f : [0, 2] \rightarrow \mathbb{R}$ satisfying $f(0) = -1$, $f(2) = 4$ and $f'(x) \leq 2$ for all $x \in [0, 2]$? Justify your answer.

Question 2: [4 marks] Suppose that $f : [0, 1] \rightarrow [0, 1]$ is differentiable function such that $|f'(x)| > 2$ for all $x \in [0, 1]$. We further assume that f' is continuous. Show that there is a unique $c \in [0, 1]$ such that $f(c) = c^2$.

Question 3: [4 marks] Let (x_n) be a sequence. Prove or disprove that if both the subsequences of even and odd indices of (x_n) converge and moreover they converge to the same limit, then (x_n) is also a convergent sequence.

Question 4: Consider the following function

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

- (a) [3 marks] Is f continuous at $(0, 0)$? Justify your answer.
- (b) [3 marks] Using the definition, find the directional derivative of f at $(0, 0)$ in the direction of unit vector $u = (u_1, u_2)$.
- (c) [3 marks] Is f differentiable at $(0, 0)$? Justify your answer.

Question 5: [4 marks] Which of the following vector fields are differentiable? Justify your answer.

- (a) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $F(x, y) = (x^2, y^3)$. If F is differentiable, find the Jacobian matrix at $(1, 2)$.
- (b) $G : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as $G(x, y, z) = (|x| + |z|, e^{xy}, \sin(xz))$. If G is differentiable, find the Jacobian matrix at $(0, 0, 0)$.

Question 6: [2 marks] Find the divergence of following vector fields:

- (a) $F(x, y, z) = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$.
- (b) $F(x, y, z) = xye^{z\hat{i}} + yze^{x\hat{k}}$.

Question 7: [4 marks] Find the curl of following vector fields:

- (i) $F(x, y, z) = -z\hat{i} + 0\hat{j} + x\hat{k}$. Is it a gradient vector field?
- (ii) $F(x, y, z) = z\hat{i} + xz\hat{j} + (xy + z)\hat{k}$. Is it a gradient vector field?

Please turn over!

Question 8: [9 marks] Solve the following.

- (i) Using double integration find the area of the region D bounded by parabolas $y = 2x^2$ and $y = 1 + x^2$.
- (ii) $\iiint_E z \, dx \, dy \, dz$, where E is enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
- (iii) $\int_C F \cdot dr$, where $F(x, y) = x^2\hat{i} + y^2\hat{j}$, and C consists of the arc of the circle $x^2 + y^2 = 4$ from $(2, 0)$ to $(0, 2)$ followed by the line segment from $(0, 2)$ to $(4, 3)$.

Question 9: [3 marks] Is the line integral $\int_C F \cdot dr$ independent of positively oriented paths having same end points, where

$$F(x, y, z) = y \cos(z)\hat{i} + x \cos(z)\hat{j} - xy \sin(z)\hat{k}?$$

Justify your answer.

Question 10: [5 marks] Verify Stokes' theorem for $F(x, y, z) = -z\hat{i} + 0\hat{j} + x\hat{k}$, where S is the hemisphere $x^2 + y^2 + z^2 = 1$, $y \geq 0$, **oriented outward**.

Question 11: [4 marks] Using Gauss divergence theorem to find $\iint_S F \cdot d\mathbf{S}$, for

$$F(x, y, z) = x^2\hat{i} + y^2\hat{j} + z^2\hat{k},$$

where S is the boundary surface of the solid $E: x^2 + y^2 + z^2 \leq 1$, lying in the first octant $x \geq 0$, $y \geq 0$, $z \geq 0$.